

Running a MedSim VRAI simulation

How to start the system, build and run a session, and change how every character, device and screen looks.

Contents

PART I · OPERATING THE SYSTEM

1. Starting the system
2. Setting up a session — the Set-up wizard
3. Printing & using the QR sheet
4. Running the session — Operate, the encounter console, PTT & intercom
5. Records, med cart & devices during the session
6. Debrief & handoff

PART II · MODIFYING CHARACTERS & IMAGES

Overview — the five places you change how things look

- 1 · Character portrait / face image
- 2 · The Avatar skins library (save once, reuse everywhere)
- 3 · The animated avatar (3D rig): turning it on and opening it
- 4 · Voice per character
- 5 · Device skins (which pump / cabinet / monitor model)
- 6 · Med-cart cabinet faces + PHI blanking
- 7 · The records system look (Helix Health EHR skin)
- 8 · Scenario Studio — generating the cast (characters, not their faces)
- 9 · Importing images via scenario import

Quick reference — where each control lives

PART III · THE EVIDENCE BASE

How the debrief & ranking are grounded

Learning-science foundations

The debriefing method

What the ranking measures — clinical judgment & competence

Standards, safety & communication

Using this as a research anchor

Key references

Operating the system

1. Starting the system

Follow the path that matches how you are running the session today. Everything happens in **Mission Control** — the card-based operator screen with three tabs across the top: **Set up**, **Operate**, and **Debrief**.

A. Running it for tablets on the classroom Wi-Fi (the normal field setup) 1. In a terminal, run the LAN launcher: `bash scripts/start.sh`. (Add `fresh` — `bash scripts/start.sh fresh` — to wipe the previous session's rooms, devices and roster and start clean. `bash scripts/start.sh stop` shuts the server down.) 2. The launcher binds the whole local network, re-mints the security certificate for the Mac's **current** Wi-Fi address, and opens Mission Control in Chrome at the LAN address (`https://<the Mac's IP>:8760`). Use that window — every QR code it prints will then carry the LAN address so tablets can reach it. 3. **Re-print the QR sheet after any launch or Wi-Fi change** — old codes may point at a previous address.

B. Running it on the Mac only (local rehearsal / building a session) 1. Run `bash scripts/run_cards.sh` from the project folder. Chrome opens Mission Control at `https://127.0.0.1:8760`. This mode is loopback-only — tablets cannot reach it, so use it for setup and dry-runs, not a live tablet session.

C. Log in - You land on the login screen, then on Mission Control's **Set up** tab. Sign in with the **Admin** seat (the master password) if you need to touch API keys, seat passwords or EHR admin; **Instructor** is enough to build and run scenarios.

D. Preflight — do this before handing anyone a QR (tablet sessions) 1. Run `bash scripts/preflight.sh`. In about ten seconds it checks the network address, that the portal is up and serving the avatar app from one origin, that the certificate covers the current address, and it regenerates a test QR. To also ping a specific tablet, pass its address: `bash scripts/preflight.sh 192.168.1.169`. 2. It ends in **PREFLIGHT PASS** (scan and go) or **PREFLIGHT FAIL** with the exact fix. Also make sure every tablet has **Private Wi-Fi Address turned OFF** — randomized addresses get dropped by the router.

E. Readiness check (inside Mission Control) - At the top of every screen is the **readiness bar** ('Checking readiness...' → a status). Click it to pop the **System status** panel: one tile per check (portal, network, certificate, voice, speech), each shown as a traffic-light plus a shape glyph (filled circle = good, triangle = caution, square = action needed), so it reads without

colour. - If the **Speech** tile is cold, click its **Warm speech model** button; it turns green on the next check. **Test all** re-runs every check. A red check is the only thing that blocks launching a scenario; amber is a caution you can proceed past.

2. Setting up a session — the Set-up wizard

On the **Set up** tab, keep **Wizard** selected (the **Board** view is the same choices laid out as cards — either can launch). Two shortcuts sit at the top right of Set up: ⚙️ **Local context** (this site's standing orders / formulary overlay) and ✨ **Generate a scenario** (Scenario Studio). Work left to right through the six numbered pills.

Optional — build a scenario first with Scenario Studio (✨ **Generate a scenario**) 1.

Describe the case: a one- or two-sentence **Premise**, **Learning objectives** (one per line), the **Acuity / arc**, **Setting**, and duration. 2. Choose the **Patient** — a new one from scratch, or one from the library. 3. Add **Local factors** (standing orders, formulary, priorities) so the cast behaves like your site. 4. Generate, review and edit the draft, then save. Saved scenarios appear in the wizard's **Scenario** step with the cast pre-filled and the chart auto-seeded.

Step 1 • Patients & rooms 1. Pick one **EHR for this session** (it applies to the whole room). 2. Set **Number of beds** — 1 for a single patient, more for a multi-patient room. Each bed gets its own join code; you pick each bed's patient in the next step.

Step 2 • Scenario 1. Pick a **scenario for each bed** — that selects the bed's patient and pre-fills its characters. 2. Optionally add **Instructor notes** (not shown to trainees).

Step 3 • Characters - Scenario characters — the cast that ships with each bed's scenario and stays at that bed. The patient is always included; uncheck any cast member to drop them. - **Shared characters** — a separate picker for universal cast (a common doctor, an allied-health team) available at **every** bed. - For each character: leave it **audio-only** (a flat portrait plus voice, runs on any tablet with a mic) or toggle **avatar** for a live 3D lip-syncing face. Avatars need a recent WebGPU-capable tablet and a larger first-load — pick them only for beds where you have that hardware. - Use the **image swatch strip** on each row to give a character a face (the one image is both the flat portrait and the 3D rig's source photo). No images yet? Click **image** to upload.

Step 4 • Assignments (roster & med-cart access) 1. Choose **Open** (default) — any student signs in at the shared terminal or cart with their name and initials and can reach every patient — or **Restrict by assignment**. 2. Under Restrict, add each student or group: **name**, **initials**, and a **role**. Nurse/student is a bedside/cart role. **Doctor, Respiratory therapist, Pharmacist** each 'fill the AI seat' — a real student plays that character. **Charge nurse, Supervisor and Instructor** automatically see all patients; you only tick patients for nurses/students and

groups. 3. Optionally click  **Generate lab reports** to produce lab/diagnostic reports for these patients to print or email.

Step 5 • Devices 1. For **each bed**, assign **basic and advanced devices** (pumps, cabinets, call bells). They are minted as QR codes at launch and appear in the cockpit ready to scan bedside. 2. Under **Common devices**, tick what the whole room shares: **Med cart** (with the option to stock each patient's medication drawer + MAR), **Medical-records terminal** (shared chart workstation), and — for multi-bed rooms — the **Nursing station** (one station monitoring every bed). A confirmation line reminds you each patient is added to the EHR on launch.

Step 6 • Review & Launch 1. Review the summary: EHR, beds and their patients, shared-cast count, device count. 2. Optionally tick **Apply local-practice overlay** so characters follow this site's standing orders / formulary on top of best practice. 3. Check the **Readiness** pill. **Launch scenario** is blocked only on a red check. 4. Click **Launch scenario**. The room comes up, devices are minted (QRs ready), and everything appears as live cards on the **Operate** tab.

3. Printing & using the QR sheet

The QR sheet is how each tablet joins the right station. Open it with  **Print QR** on the Operate tab's room-controls bar (or the  link from any bed's console), then use your browser's Print to send it to paper or PDF. **Re-print after every launch or Wi-Fi change** so the codes carry the current address.

One page per patient (bed), leading with: -  **Patient avatar (animated rig)** — scan on the bedside tablet to launch that patient's animated 3D face rig. (Beds set to audio-only won't have a rig; they use the character voice stations instead.) -  **Chat station** — a bedside student joins to talk to the patient. -  **EHR station** — opens that patient's chart, notes and orders in the session's records system. -  **Device station** — joins this bed's IV pumps, cabinets and call bells. -  **Scenario characters** — one small QR per this bed's own cast (family, etc.), each marked avatar (animated) or voice. -  **Devices** — one QR per bound device, so you can stick each code on the actual hardware.

A final 'Common' page carries what the whole room shares: -  **Common characters** — the shared cast; each is **one tablet, all beds** (see running the session below). -  **Nursing Station** — the supervisor view that monitors every bed (uses the **room code**, not a bed code); it is not repeated on each patient page. -  **Med cart** — the shared dispensing cart reaching every patient's MAR drawer. -  **Medical Records Terminal** — any student signs in with name + initials to read any patient's chart (uses the room code; no supervisor needed).

4. Running the session — Operate, the encounter console, PTT & intercom

The Operate cockpit 1. Click the **Operate** tab. The **System status** collapses to the small readiness button; click it any time for the tiles. 2. **Operations cards** fill the panel — one per **bed/patient** (with join code and device/character counts), one per **shared character**, one per **Med cart**, the **Nursing station**, and a  **Network & devices** card that opens a live who's-connected map. 3. Every card has **Open**  (open in place) and **Pop out**  (a new window you can drag to a second monitor and leave running live); some also show a **QR** link. 4. The **room-controls bar** runs the whole room: ► **Start all** (arms a configured room), || **Pause all** / ► **Resume all** (gate student input), ⚡ **Inject scene**,  **Print QR**, and ■ **End all (debrief)**. 5. ⚡ **Inject scene** opens a small dialog: pick a **Scene** (e.g. vitals.drop hypotensive crash, pump.alarm IV occlusion, code.blue full arrest, note.instructor free-text), a **Target** (all beds), optional JSON params, then ⚡ **Inject**.

The encounter console (drill into one patient) - Open (or Pop out) a patient card to reach that bed's console. ← **Back to Mission Control** returns you to Operate. Per-bed controls ► **Start** / || **Pause** / ■ **End** sit top-right (a per-bed End does **not** fire the cohort debrief — that only happens on the room's End all). - The console shows live telemetry, an ECG strip, ventilator state, the device list, medications, the bed's QR codes, live transcript, connected stations, telemetry overrides, a per-bed scene injector, staged medication errors, and documents.

Push-to-talk (PTT) — play a character by voice 1. In  **Characters · voices · engage**, pick an ElevenLabs voice per character; ► **Test** previews it.  **Engage** opens a text chat to play that character. 2. In  **Engage a character (push-to-talk)**, click a character chip, then **hold the button and speak**; on release the character replies in its assigned voice and the turn is logged. The first press prompts for the microphone — click Allow. 3. Where the mic isn't available (e.g. Safari), **type a line and press Enter** instead — same result.

Shared characters — one tablet, many patients - A shared character (the common doctor) runs from its **single** QR: on that one tablet, ask about Bed 1 and it answers about Bed 1; ask about Bed 2 and it answers about Bed 2. PTT and the type fallback both work on the shared-character page.

Intercom (live staff audio) - On the **Nursing Station**, each bed card has a  **Hold to talk** button and the header has  **Page all rooms** — hold to speak live to one bed or to every bed at once; a typed fallback per bed is also there. - At the bedside, students use  **Call nurses station** to talk back, and a banner lights up when the station pages them. (Intercom needs HTTPS and a first tap to grant the mic; the tablets must be able to reach each other on the Wi-Fi.)

5. Records, med cart & devices during the session

Medical records - Shared records terminal: a student scans the **Medical Records Terminal** QR, signs in with **name + initials**, and sees only the patients they're cleared for (everyone under Open access; their assignments under Restrict). Every chart-open is tagged with that name + initials. - **From the operator side:** in a patient's encounter console,  **Open EHR** opens that bed's full chart, notes, vitals and order entry in a new window (or **Open EHR on this device** from the QR panel to skip the scan). - **Supervisor records:** on the Nursing Station,  **Open Supervisor Records** prompts for the supervisor's 2 initials and opens the records workstation with admin powers (add lab, add note).

Med cart - A student scans the **Med cart** QR and meets a BD-Pyxis-style cabinet face. It is a shared terminal: each student **signs in** (with initials), the cart shows only the patients that student may reach, and they **pick the patient**, open the **MAR drawer**, and pull/scan the medication. The cart auto-locks after inactivity so it returns clean for the next student. Only meds you ticked as active in a patient's **Medications** card appear on that patient's cart (if you never touched it, all meds show). - Room-wide dispensing runs through the **Med cart** card on Operate; per-patient medications, staged errors and handoff are managed from that patient's encounter console, not a session-wide page.

Devices - Bedside devices (IV pumps, cabinets, call bell, bed alarm, code blue, fire alarm) join via the bed's **Device station** QR or a device's own QR. In the encounter console, the **Devices** card lists what's bound with its state, and  **Managed devices** opens the full device + medication manager for that bed in a modal. - The  **Network & devices** card on Operate opens a live map of who is connected and how devices are linked; you can toggle the layout, and (with **Manage links** ticked) click a device to cycle it available → fault → auto for the session. These overrides are session-scoped and clear when the session ends.

Nursing Station (monitoring) - The Nursing Station shows an **Alarms** board (highest severity first), per-bed cards, and an **Alarm thresholds** panel where the supervisor sets low/high bounds per vital (HR, SpO₂, RR, BP) and marks which ECG rhythms count as dangerous. Breaches surface on the alarm board.

6. Debrief & handoff

Handoff (during the session, per patient) - In a patient's encounter console, the  **Shift handoff** card runs a structured handoff for that bed. Pick a **Mode** — the student **gives** report (the AI receives and probes gaps) or **receives** report (the AI gives it) — choose the AI counterpart (charge nurse / oncoming), then ► **Start handoff**. Engage the counterpart by PTT, then **Score** shows the SBAR coverage grid (check/cross) and the student's self-rating vs the measured result.

Ending the session 1. On Operate's room-controls bar, click ■ **End all (debrief)** and confirm. This ends every bed, disconnects connected students, and **builds the cohort debrief**, then opens it. (A per-bed End alone does not save the cohort debrief.) 2. The **Debrief** tab links to **Open debriefs** for reviewing past runs.

Reviewing the cohort debrief - The cohort debrief opens in the PEARLS structure across tabs: **Reactions** (with an instructor notes box you Save), **Description** (key facts), **Analysis** (a performance-frame table per encounter — duration, turns, clinical-judgment coverage — plus cohort-wide character engagement), **Application** (add each student's commitments for next shift), **Per-encounter** (expand any bed for its session details and transcript), and **Summary** (encounters, students, total turns, durations). - Use 🖨 **Print / Save as PDF** to hand off or archive the debrief. ← **All cohorts** returns to the list of past cohort debriefs.

Handoff between operators / resuming after a restart - If the server is restarted gracefully, the live room resumes on its own. On the Operate tab a **Resume session** banner appears — click it to restore the room (scenario, beds, devices, shared cast) intact. After any restart, **re-login** so characters respond in-character rather than echoing. Launching a **fresh** scenario clears a resumed one — a new launch is not a resume.

Modifying characters & images

Overview — the five places you change how things look

Everything visual in a session is set from the operator side before or during a run. There are five surfaces where you change appearance, and this section walks each one:

1. **Character faces (portraits & avatars)** — the image a character wears, chosen in the **Launch Wizard**, on the **Scenarios** page, or on the **Persona library** page.
2. **The Avatar skins library** (**image / Avatar skins** page) — your reusable library of saved faces.
3. **Voices** — the spoken voice each character uses, on the **Character voices** panel.
4. **Device skins** — which IV pump / cabinet / monitor brand a device looks like, in the **Add device** dialog.
5. **The records system look (Helix Health / Cyrus Care / Meridian EHR)**, chosen once per session in the wizard.

A useful rule to remember: **one image per character serves two purposes** — it is both the flat portrait a student sees on an audio-only tablet *and* the source photo for the animated 3D face. You never upload a face twice.

1 • Character portrait / face image

A character's face comes from one image. There are three ways to give a character that image, and three screens where you can do it.

Where to set it - Launch Wizard → step 3 (Characters): each character row has a small strip of **image swatches** (little square thumbnails). Click a swatch to give that character that face; click the **∅ (No image)** swatch to remove it. The change saves immediately. - **Scenarios page:** each character in a scenario shows a  **avatar assigned** or  **no avatar yet** badge and its face controls. - **Persona library page:** each persona card has a  **Use avatar face: on/off** row with the same swatch strip.

The three sources of a face 1. **Upload your own photo (a "skin").** If you have a consented photo, save it once on the **Avatar skins** page (see section 2), then pick it from the swatch strip. In the wizard, if you have no images yet, the row shows a **image** link that opens the Avatar skins page in a new tab — upload there, then reopen the wizard and the swatch

appears. 2. **AI-generate the face ("🌟🌟 Request from FACE ENGINE")**. On the **Persona library** page and the **Scenarios** page, any character with no face yet shows a 🌟🌟 **Request from FACE ENGINE** (or 🌟🌟 **FACE ENGINE**) button. Click it and the system generates a fully-synthetic portrait for that character (takes a few seconds — it only needs the character's age range and sex). This requires the FACE ENGINE service to be running; if it is not reachable you get a plain error message and nothing changes. 3. **The placeholder (default)**. A character you never give a face to renders a neutral slate-gray head silhouette. Nothing breaks — the voice and rig still work — it just has no real face.

AI disclosure. Any portrait made by FACE ENGINE carries an **AI-generated** tag (a small badge, tooltip: "This portrait is fully synthetic — not a real person"). This appears on the Persona library and Scenarios pages so it is always clear which faces are synthetic. Uploaded photos and the placeholder do not get this badge.

Accepted formats & size (for uploaded faces): PNG, JPEG, or WEBP; up to **8 MB** per image (see section 2). AI-generated faces are produced as PNG automatically.

2 · The Avatar skins library (save once, reuse everywhere)

The **Avatar skins** page (left nav: **Avatar skins**, or the **image** links elsewhere) is your reusable face library. Save a face once with a label and then assign it to any character or persona without re-uploading.

To save a face 1. Go to the **Avatar skins** page. 2. Type a **Skin label** (e.g. "Tired night-shift RN") so you can find it later. 3. Click **Choose file** and pick the image. **Accepted types: PNG, JPEG, or WEBP. Maximum size: 8 MB** (a larger file is rejected with "Image too large (max 8 MB)"). 4. Click  **Save skin**. It appears as a thumbnail card in your library.

To assign a saved face to a character - On each skin card, use the **— assign to... —** dropdown to pick a character, then click **Assign**. That character now shows  **avatar assigned** on the Scenarios page, and that is the face the tablet renders. (You can also assign from the wizard/Persona/Scenarios swatch strips described in section 1 — they read from this same library.)

To remove a face: click **Delete** on the skin card (it asks you to confirm).

Note: assigning copies the image to that character, so deleting the library skin later does not blank an already-assigned character.

3 · The animated avatar (3D rig): turning it on and opening it

Giving a character an image is separate from making that character an **animated 3D face** rather than a flat portrait with voice.

Audio-only vs. animated (set in the wizard). In **Launch Wizard → step 3 (Characters)**, each character has an **avatar** checkbox: - **Left unchecked (default):** the character is **audio-only** — a flat portrait plus voice and push-to-talk. Runs on any low-cost tablet with a browser and microphone. - **Checked:** the character gets the **animated 3D rig** — a live face that lip-syncs. The wizard's ⓘ **Avatar hardware & cost** note warns this needs a **WebGPU-capable tablet** (a recent iPad or high-end Android) and a larger first-load download (about 100 MB, cached after the first open). Patients default to animated; other roles default to audio-only.

Opening a character's avatar on a tablet (via QR). - Print the **QR sheet** for the session: each patient page **leads** with a ⓘ **Patient avatar (animated rig)** QR, and each scenario/shared character has its own QR labeled **avatar (animated)** or **voice**. Scan the QR on the tablet and the face opens full-screen, bound to that character. - From the operator control view you can also open the **Pair tablet / Assign avatar to a tablet** page for a character (the ⓘ **Develop avatar** → link opens the live face app for checking it). These pages show the character's name, whether it has a **custom portrait** ✓ or the **placeholder**, and the QR/URL to scan.

Face transparency (opacity / ghost tint). When an avatar is launched it opens at a default translucency (mid-stop, ~0.66) so it can overlay a manikin. This is a launch setting, not a per-character library edit; the pairing/launch URL carries an **opacity** value, and a per-scenario or per-character **ghost color** tint can be set in the scenario data (defaults to clinical white). A 🐛 **Debug QR** toggle on the launch page hands the tablet a build with an on-screen **translucency slider** for fine-tuning during setup.

4 • Voice per character

Each character speaks in an assigned voice, set on the **Character voices** panel in the operator control view.

To change a character's voice 1. Open the **Character voices** panel (in the ops/control view; also reachable as **Characters • voices** in the field flow). 2. Each character has a **voice dropdown**. The system pre-ranks about **5 candidate voices** matched to that character's sex and age band; the list also always includes a **Browser voice (fallback)** option at the top. 3. Pick a voice from the dropdown — it saves immediately ("✓ assigned"). 4. Click ► **Test / Preview** to hear a sample line in that voice before committing.

Two voice engines. A health badge at the top of the panel tells you which is active: - ● **ElevenLabs live** — high-quality synthesized voices (this is what the ranked candidates are). Preview and in-session speech use it. - ● **Browser voices (ElevenLabs not configured)** — if

no ElevenLabs key is set, or a character is left on **Browser voice (fallback)**, speech uses the tablet/computer's built-in text-to-speech instead. The system automatically degrades to the browser voice if ElevenLabs is briefly unavailable.

Once assigned, push-to-talk and typed replies both come back **spoken in that character's chosen voice** (not the browser default).

Character text personality (register, sentence length, example lines, "never says") is edited separately on the **New/Edit character** form under the **Voice** section — that shapes **what** the character says, while the Character voices panel sets **how it sounds**.

5 • Device skins (which pump / cabinet / monitor model)

The on-screen look of a simulated device — its make and model — is chosen when you create the device, in the **Add device** dialog (operator control view → **Add device**; or the wizard's **Devices** step, which mints device QRs per bed).

To pick a device's brand/model 1. Click **Add device**. 2. Choose a **Kind** — the dropdown is grouped into **Basic devices** (IV pump, enteral pump, dispensing cabinet, patient alarm) and **Advanced devices** (telemetry monitor, vent monitor, ventilator). 3. Choose a **Model / brand** — the model dropdown updates to the skins available for that kind. For example: - **IV pump**: BD Alaris (plus additional pump skins such as Plum 360, Spectrum IQ, Infusomat Space, and others that ship in the library). - **Enteral pump**: Kangaroo OMNI (and Kangaroo Joey). - **Dispensing cabinet**: five cabinet faces — **BD Pyxis MedStation**, Omnicell XT, Arxium, Capsa, and Touchpoint. - **Advanced**: bedside telemetry monitor, ventilator display, ventilator (controls). 4. Give it a **Label** (e.g. "Bed 3 IV"), optionally assign it to a character/patient, and click **Mint QR**. Scan that QR on the device tablet and it renders that model's skin.

To re-skin an existing device, open its card's **Detail / inject** — you can reassign the character there; to change the model itself, mint a new device with the model you want. Each device's card shows its current kind and model (e.g. "IV pump · BD Alaris").

6 • Med-cart cabinet faces + PHI blanking

The medication cart is a room-level cabinet, created on the **Multi-Patient Control** (room) page under the  **Med carts** panel.

Creating / linking a cart 1. In the  **Med carts** panel, type a **Cart label** (e.g. "ICU Cart A"). 2. Tick which beds/encounters the cart serves (✓ **All beds** / ✗ **None** shortcuts help), then **+ Add med cart**. The cart gets its own QR and shows every linked patient's meds grouped by patient name.

Choosing the cabinet face. Carts render as a dispensing cabinet and default to the **BD Pyxis** face. To present a specific cabinet brand, add the cabinet as a device via **Add device → Kind: Dispensing cabinet → Model / brand**, where all **five cabinet skins** are available: **BD Pyxis MedStation, Omnicell XT, Arxium, Capsa, Touchpoint**. (Note: inside an encounter console's device modal the cabinet option is hidden on purpose — carts are managed at the room level, not per bed.)

Baked PHI blanking (automatic, not a setting you toggle). The cabinet skin art ships with a placeholder patient name baked into it (e.g. "MARTIN, ELENA"). For HIPAA hygiene the cabinet **screen never displays that baked name** — it is blanked out automatically. A real patient name appears only inside the MAR overlay, and only once the scenario is running and a patient has been selected. There is nothing you need to configure; the blanking is enforced by the system.

7 • The records system look (Helix Health EHR skin)

The electronic health record students chart in has a recognizable branded look, and you pick which one **once per session**.

Where to choose it. In the **Launch Wizard → Devices step**, the field **EHR for this session** is a dropdown. (When you build encounters on the multi-patient control page, each bed also has an **EHR** dropdown, defaulting to the first option.)

The three EHR skins - Helix Health — Epic-style, cobalt + mustard. Tabs: Chart, Vitals, Notes (SmartPhrase), Ambient Scribe, CPOE, Results, MAR. - **Cyrus Care** — Cerner-style, teal/navy. Tabs: Worklist, Patient Summary, iView Flowsheet, PowerNote, Ambient Capture, PowerOrders, Results Review. - **Meridian EHR** — Meditech-style, sage/cream. Tabs: My Day, Chart, Vitals, Document, Voice+Scribe, Orders, Results.

Whichever you pick brands the student-facing Records Terminal (the masthead reads, e.g., **Helix Health • Electronic Health Record**) and sets its tab layout and order catalog. It is **one EHR for the whole session** — you cannot mix brands across beds in a single run. This is a fixed choice of three; the branding text and colors are not free-text editable from the operator UI.

8 • Scenario Studio — generating the cast (characters, not their faces)

Scenario Studio (🌟 **Scenario Studio**, reached from **Set up**) uses AI to draft a whole runnable scenario from a plain-English description — and that draft **includes the supporting cast** (characters).

How it makes characters 1. Fill in **The case** (premise, objectives, acuity, setting, duration), the **Patient** (new from scratch or picked from the library), **Local factors**, and a **Supporting cast** hint (e.g. "attending, charge nurse, respiratory therapist, anxious spouse"). 2. Click **Generate draft**. The **Review & edit** panel opens with a **Supporting cast** list you can edit, add to (+ add character), or remove — nothing is saved until you click **Save to scenario library**. 3. Saved scenarios then appear in the **Launch Wizard's Scenario step** with the cast pre-filled.

Important scope limit: Scenario Studio generates the characters as records (name, role, behavior) — it does **not** generate their face images or avatars. After saving, give each character a face the normal way: pick a saved skin, or click ✨ **Request from FACE ENGINE** on the Scenarios / Persona library page (section 1). So the flow is: Studio drafts *who* is in the room; sections 1–2 decide *what they look like*.

9 · Importing images via scenario import

You can bring a scenario in from another MedSim VRAI system via **Scenarios → Import** (the **Import a scenario** page), which accepts a `.medsim-scenario.json` bundle.

What travels and what does not. The bundle carries the scenario, its characters, and support documents — but **face images (portraits and skins) are deliberately NOT included in the file**. Faces are consent-gated and tied to the system they live on, so they do not travel between systems.

What you do after importing 1. On the **Import a scenario** page, choose the `.json` bundle and click **Import**. Imports never overwrite — a scenario or character whose id already exists lands under a new id (identical characters are simply linked). 2. Read the **Before first launch** checklist the import shows. If any imported character had a face on the source system, the checklist tells you to **"Assign an avatar for '<character>' (portraits don't travel — consent is per-system)."** 3. Click **Review & edit**, then give those characters faces the normal way — assign a saved skin or ✨ **Request from FACE ENGINE** (section 1).

In short: scenario import moves the cast and the case, but you always (re)attach faces on the importing system. It is not a way to bulk-import images themselves — the only image upload path is the **Avatar skins** page (section 2).

Quick reference — where each control lives

| What you want to change | Screen | Control | |---|---|---| | Give a character a face (upload) | Avatar skins page, then a swatch strip | 📁 **Save skin** → **Assign** / image swatch | | Give a character a face (AI) | Persona library or Scenarios page | ✨ **Request from FACE ENGINE** | | Remove a character's face | Any swatch strip | ∅ (**No image**) swatch | | Flat portrait vs.

animated 3D face | Launch Wizard → Characters (step 3) | **avatar** checkbox | | Open a face on a tablet | QR sheet / Pair-tablet page | scan the character's  **avatar** QR | | Change a character's voice | Character voices panel | voice **dropdown** + ► **Test** | | Change a device's brand/model | Add device dialog | **Kind** → **Model / brand** | | Cabinet face (5 options) | Add device → Dispensing cabinet | **Model / brand** dropdown | | Records system look (3 options) | Launch Wizard → Devices step | **EHR for this session** dropdown | | Draft a new cast with AI | Scenario Studio | **Generate draft** | | Bring in a scenario | Scenarios → Import | upload `.medsim-scenario.json` (faces re-assigned after) |

Formats & limits worth repeating: uploaded faces = **PNG / JPEG / WEBP, 8 MB max**; import bundles = `.json` ; animated avatars need a **WebGPU tablet** (~100 MB first load); AI-generated faces carry an **AI-generated** disclosure badge.

The evidence base — standards & science

How the debrief & ranking are grounded

Nothing in the debrief or the performance ranking is invented for this product. Each moving part is a recognised, evidence-based practice from healthcare-simulation education, adopted deliberately. Instructors and students can see exactly which standard each feature rests on, and researchers can treat this table as the map from a system behaviour to its literature.

| System feature | Grounded in | Anchor reference | |---|---|---| | Debrief structure (Reactions · Description · Analysis · Application · Summary) | **PEARLS** blended debriefing | Eppich & Cheng (2015) | | Facilitator stance (curious, non-shaming inquiry) | **Debriefing with Good Judgment** / advocacy-inquiry | Rudolph et al. (2006, 2007) | | Reflective "commitments for next shift" | **Experiential learning** + reflective practice | Kolb (1984); Schön (1983) | | Handoff coverage grid | **SBAR** structured communication | IHI / Joint Commission | | "Clinical-judgment coverage" per encounter | **Tanner Clinical Judgment Model** / NCSBN NCJMM | Tanner (2006); NCSBN (2019) | | Repeated, feedback-rich runs | **Deliberate practice** / mastery learning | Ericsson (2004); McGaghie et al. (2011) | | Overall design & conduct | **INACSL Standards of Best Practice** | INACSL (2021) |

The ranking is a **structured, criterion-referenced** judgement, not a black-box score: the criteria are visible in the debrief, editable by the instructor, and recorded for review.

Learning-science foundations

Experiential learning (why simulation + debrief works)

A simulation is a **concrete experience**; the debrief is the **reflective observation** and **abstract conceptualisation** that turn it into transferable learning, and the "commitments for next shift" are the **active experimentation** that closes Kolb's cycle (Kolb, 1984; building on Dewey and Lewin). Schön's **reflection-in-action** / **reflection-on-action** (1983) is why the debrief separates what was noticed in the moment from what is understood afterward.

Deliberate practice & mastery learning

Skill is built by **repeated, focused practice with immediate, specific feedback against a standard** (Ericsson, 2004), and simulation-based **mastery learning** — practising to a defined performance bar rather than a fixed seat-time — produces measurable downstream clinical

improvements (McGaghie et al., 2011). The system supports this by making runs cheap to repeat and every run debriefed against the same rubric.

The debriefing method

PEARLS — the structure the debrief tabs follow

The cohort debrief is organised as **Reactions → Description → Analysis → Application → Summary**, the **PEARLS** (Promoting Excellence and Reflective Learning in Simulation) framework (Eppich & Cheng, 2015), a blended approach that combines plus/delta, focused facilitation, and directive feedback within one flow.

Debriefing with Good Judgment

The facilitator stance the guide encourages — stating an observation, then asking about the trainee's frame — is the **advocacy–inquiry** method of "debriefing with good judgment" (Rudolph et al., 2006, 2007): rigorous about performance while genuinely curious about the reasoning behind it, which preserves psychological safety. Fanning & Gaba (2007) review why this reflective conversation, not the scenario alone, is where learning consolidates.

Judging the debrief itself

Debrief *quality* is itself measurable: the **DASH** (Debriefing Assessment for Simulation in Healthcare; Brett-Fleegler et al., 2012) gives programmes a validated instrument to rate and improve facilitation — useful for faculty development and for research on facilitator effects.

What the ranking measures — clinical judgment & competence

Clinical judgment

The per-encounter "**clinical-judgment coverage**" the analysis reports maps to Tanner's **Clinical Judgment Model** — *noticing, interpreting, responding, reflecting* (Tanner, 2006) — the same construct the **NCSBN Clinical Judgment Measurement Model (NCJMM)** operationalises for the Next Generation NCLEX (NCSBN, 2019). Progression across encounters is read through Benner's **novice-to-expert** trajectory (Benner, 1984).

Levels of assessment

Simulation assesses the upper tiers of **Miller's pyramid** — "shows how" and "does," not just "knows" (Miller, 1990) — and outcomes can be framed at **Kirkpatrick's** four levels (reaction, learning, behaviour, results; Kirkpatrick & Kirkpatrick, 2006).

Validated rubrics the approach draws on

Criterion-referenced scoring of simulated performance has established, published instruments the system's rubric is designed to be compatible with: the **Lasater Clinical Judgment Rubric** (Lasater, 2007, operationalising Tanner) and the **Creighton Simulation Evaluation Instrument (C-SEI, later revised as the C-CEI)** (Todd et al., 2008). Instructors can align the editable rubric to these, or to their program's own competency map (e.g. QSEN).

Standards, safety & communication

The system is built to sit inside recognised professional standards, so a program can defend its use in a curriculum:

- **INACSL Healthcare Simulation Standards of Best Practice** (2021) — the field standards for simulation design, facilitation, *debriefing*, evaluation, and professional integrity.
- **NLN/Jeffries Simulation Theory** (Jeffries, 2005) — the design framework relating facilitator, participant, educational practices, and outcomes.
- **QSEN** competencies (Cronenwett et al., 2007) — patient safety, teamwork, informatics, and quality, which scenarios and the rubric can target.
- **SBAR** (IHI; endorsed in Joint Commission handoff expectations) — the structured Situation-Background-Assessment-Recommendation format the handoff coverage grid scores.

Using this as a research anchor

Because every run is debriefed against a visible, editable rubric and recorded, the system produces a **structured, timestamped, criterion-referenced dataset** — per-encounter performance frames, SBAR coverage, clinical-judgment coverage, character-engagement metrics, and each student's reflective commitments — that a research project can build on, modify, or validate.

Ways to build on it

- **Instrument validity/reliability:** compare the system's AI-assisted clinical-judgment coverage against a human-scored Lasater CJR or C-SEI/C-CEI on the same encounters.
- **Outcomes:** relate debrief-derived measures to NGN/NCJMM readiness or to downstream clinical performance (mastery-learning translational outcomes).
- **Facilitation:** use DASH to study facilitator effects on learning.
- **Dosing:** test deliberate-practice repetition (how many runs to a mastery bar).

To anchor a project

Define your construct and instrument, map them to the references below, configure the rubric to match, export the run data, and pre-register the analysis. Cite this section's framework set as the theoretical grounding and extend from there — the rubric and scenario library are meant to be modified for a study, not treated as fixed.

Key references

A curated foundation — the works the debrief and ranking rest on — not an exhaustive review. Confirm the current edition of each standard (INACSL and the NCSBN NCJMM are revised periodically).

Learning science

1. Kolb, D. A. (1984). *Experiential Learning: Experience as the Source of Learning and Development*. Prentice Hall.
2. Schön, D. A. (1983). *The Reflective Practitioner: How Professionals Think in Action*. Basic Books.
3. Ericsson, K. A. (2004). Deliberate practice and the acquisition and maintenance of expert performance in medicine and related domains. *Academic Medicine*, 79(10 Suppl), S70–S81.
4. McGaghie, W. C., Issenberg, S. B., Cohen, E. R., Barsuk, J. H., & Wayne, D. B. (2011). Does simulation-based medical education with deliberate practice yield better results than traditional clinical education? A meta-analytic comparative review of the evidence. *Academic Medicine*, 86(6), 706–711.

Debriefing

1. Fanning, R. M., & Gaba, D. M. (2007). The role of debriefing in simulation-based learning. *Simulation in Healthcare*, 2(2), 115–125.
2. Rudolph, J. W., Simon, R., Dufresne, R. L., & Raemer, D. B. (2006). There's no such thing as "nonjudgmental" debriefing: A theory and method for debriefing with good judgment. *Simulation in Healthcare*, 1(1), 49–55.
3. Rudolph, J. W., Simon, R., Rivard, P., Dufresne, R. L., & Raemer, D. B. (2007). Debriefing with good judgment: Combining rigorous feedback with genuine inquiry. *Anesthesiology Clinics*, 25(2), 361–376.
4. Eppich, W., & Cheng, A. (2015). Promoting Excellence and Reflective Learning in Simulation (PEARLS). *Simulation in Healthcare*, 10(2), 106–115.
5. Brett-Fleegler, M., et al. (2012). Debriefing Assessment for Simulation in Healthcare (DASH): Development and psychometric properties. *Simulation in Healthcare*, 7(5), 288–294.

Clinical judgment & competency assessment

1. Tanner, C. A. (2006). Thinking like a nurse: A research-based model of clinical judgment in nursing. *Journal of Nursing Education*, 45(6), 204–211.
2. Benner, P. (1984). *From Novice to Expert: Excellence and Power in Clinical Nursing Practice*. Addison-Wesley.
3. Lasater, K. (2007). Clinical judgment development: Using simulation to create an assessment rubric. *Journal of Nursing Education*, 46(11), 496–503.
4. National Council of State Boards of Nursing. (2019). *NCSBN Clinical Judgment Measurement Model* (Next Generation NCLEX). NCSBN.
5. Miller, G. E. (1990). The assessment of clinical skills/competence/performance. *Academic Medicine*, 65(9 Suppl), S63–S67.
6. Kirkpatrick, D. L., & Kirkpatrick, J. D. (2006). *Evaluating Training Programs: The Four Levels* (3rd ed.). Berrett-Koehler.
7. Todd, M., Manz, J. A., Hawkins, K. S., Parsons, M. E., & Hercinger, M. (2008). The development of a quantitative evaluation tool for simulations in nursing education (the Creighton Simulation Evaluation Instrument, C-SEI; later revised as the C-CEI). *International Journal of Nursing Education Scholarship*, 5(1), Article 41.

Standards, safety & communication

1. INACSL Standards Committee. (2021). Healthcare Simulation Standards of Best Practice. *Clinical Simulation in Nursing* (standards series).
2. Jeffries, P. R. (2005). A framework for designing, implementing, and evaluating simulations used as teaching strategies in nursing. *Nursing Education Perspectives*, 26(2), 96–103.
3. Cronenwett, L., et al. (2007). Quality and safety education for nurses. *Nursing Outlook*, 55(3), 122–131.
4. Institute for Healthcare Improvement. *SBAR Tool: Situation-Background-Assessment-Recommendation* (structured handoff communication).